Bond-angular order in two-dimensional Lennard-Jones and hard-disk systems

Katherine J. Strandburg,* John A. Zollweg, and G. V. Chester
Laboratory of Atomic and Solid State Physics, Cornell University, Ithaca, New York 14853
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As a new probe of the nature of the two-dimensional melting transition, a Monte Carlo study of nearest-neighbor bond-angular susceptibility on various length scales has been performed. This probe has been used to distinguish between a hexatic phase and a two-phase region. We conclude that the melting transition of both the low-density Lennard-Jones potential system and the hard-disk system is first order in these finite simulations. No qualitative difference in behavior between the two systems was observed.

The nature of the two-dimensional melting transition remains a matter of controversy despite numerous computer simulations[1] and experiments[1,2] designed to explore the issue. Part of the controversy has revolved around the identification of the intermediate region observed between the solid and fluid in constant-density simulations. While many authors have concluded that it is a region of two-phase coexistence, many others have concluded either that it is a hexatic phase, or at least that the hexatic interpretation cannot be ruled out. The low-density Lennard-Jones system has been one system for which such a disagreement has remained.[3] The most compelling evidence for a first-order transition has come from constant-pressure simulations.[4] However, the correctness of these methods in treating phase transitions has recently been called into question.[5]

In this paper we report results of simulations of two-dimensional Lennard-Jones and hard-disk systems in which we probe bond-angular order on various length scales. Our data are entirely consistent with the hypothesis of two-phase coexistence and inconsistent with the existence of a hexatic phase. We find no qualitative difference between the low-density Lennard-Jones and the hard-disk systems in the behavior of the angular order.

Rather than attempt to look for a slow asymptotic decay in the orientational correlation function $g_6(r)$ (which would almost certainly be obscured by finite-size effects), we have, following a suggestion by Tobochnik,[6] computed the angular susceptibility χ_6. We define

$$\chi_6=\left\langle \left| \frac{1}{N}\sum_l \frac{1}{n_l}\sum_j e^{6i\theta_{lj}} \right|^2 \right\rangle ,$$

where the sum on l is over all particles, the sum on j is over nearest neighbors, n_l is the number of nearest neighbors of particle l, and N is the number of particles in the system. The nearest-neighbor bond angle θ_{lj} is the angle which the bond between particles l and j makes with an arbitrary axis. We have used this quantity to examine the system on different length scales by dividing our system

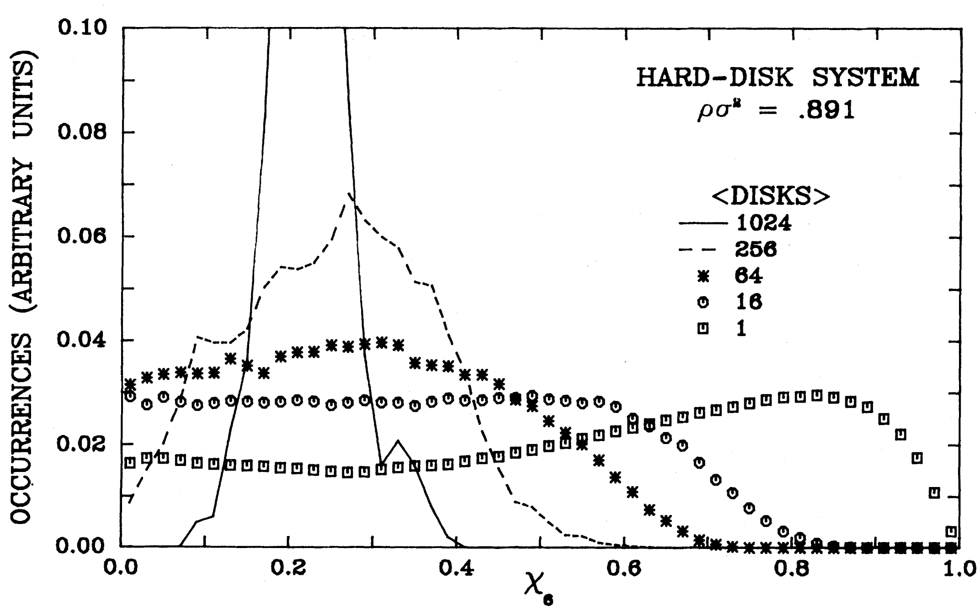

FIG. 1. Distribution of occurrences of the bond-angular susceptibility χ_6 for hard disks at $\rho\sigma^2=0.891$. Data are displayed for subsystems containing an average of 1024, 256, 64, 16, and 1 particles.

of 1024 particles into subsystems containing an average of 256, 64, 16, and 4 particles. χ_6 is then computed for each subsystem. In the solid χ_6 will be large due to the presence of long-range order.[3] In the fluid χ_6 will be small (of order $1/N$). If the system exhibits an inhomogeneous two-phase region, then we would expect that at sufficiently small length scales the distribution of values of χ_6 obtained could be modeled by a combination of solid and fluid distributions. On the other hand, if the system exhibits a homogeneous hexatic phase, then varying the size of the subsystems should not lead to any qualitative change in the distribution of values of χ_6. To confirm our expectations about the behavior of χ_6 in a hexatic phase we performed a subsystem analysis of the susceptibility of a two-dimensional system of XY spins in the low-temperature critical phase (the spin ordering of which is analogous to the orientational order in a hexatic phase).

A typical Monte Carlo run consisted of 50 000 to 100 000 passes. The angular susceptibility χ_6 was computed every 10 or 20 passes. We defined our subsystems simply by dropping a rectangular grid over our system. The fluctuations in the mean number of particles in any subsystem were negligible in the solid phase and of order $(N_s)^{1/2}$ in the fluid phase, where N_s is the average number of particles per subsystem. The fluctuations in the average value of χ_6 computed for the entire system were large in the intermediate region. However, the fluctuations in the shapes of the distributions of χ_6 for the subsystems were found to be small.

Figure 1 shows the distribution of values of χ_6 on several length scales for the hard-disk system at a point near the middle of the intermediate region ($\rho\sigma^2=0.891$). In Fig. 2 the results for a low-density ($\rho\sigma^2=0.856$) Lennard-Jones system near the middle of the intermediate region ($T^*=0.77$) are shown for comparison. The similarity is striking. This qualitative agreement persists throughout the intermediate region and in the solid and fluid phases. Indeed, the average values of χ_6 for the total system at melting and freezing are essentially the same for the two systems. Note that as the subsystem size is decreased the distribution broadens and becomes rather flat but does not shift substantially. There is considerable contribution from subsystems with small fluidlike values of χ_6, as well as from subsystems with higher solidlike values.

This inhomogeneous behavior may be contrasted with that of the two-dimensional triangular XY model. The distributions of spin susceptibility for subsystems of 256, 64, 16, and 4 spins at $T^*=1.2$ (approximately 20% below T_c^*) are shown in Fig. 3. This temperature is in the low-temperature critical region but is well above the spin-wave regime. The behavior displayed in Fig. 3 is markedly different from that shown in Figs. 1 and 2. No contributions from low values of χ_6 are seen on any length scale. The peak shifts fairly substantially and broadens as the size of the subsystem is decreased, but does not change its shape. Similar, although slightly sharper, behavior was observed at lower temperatures. The high-temperature phase of the XY model displayed distributions peaked near zero with fairly long tails similar to the fluid away from freezing.

We conclude from these results that the Lennard-Jones and hard-disk systems show essentially the same behavior. This behavior is consistent with the idea of an inhomogeneous two-phase coexistence region and inconsistent with the type of homogeneous behavior observed for the critical phase of the XY model.

In order to test the hypothesis of two-phase coexistence more rigorously we undertook to model the distributions in the intermediate region as combinations of the distributions in the liquid and solid. It is a simple matter to determine the proportions of fluid and solid expected at any density in the coexistence region.

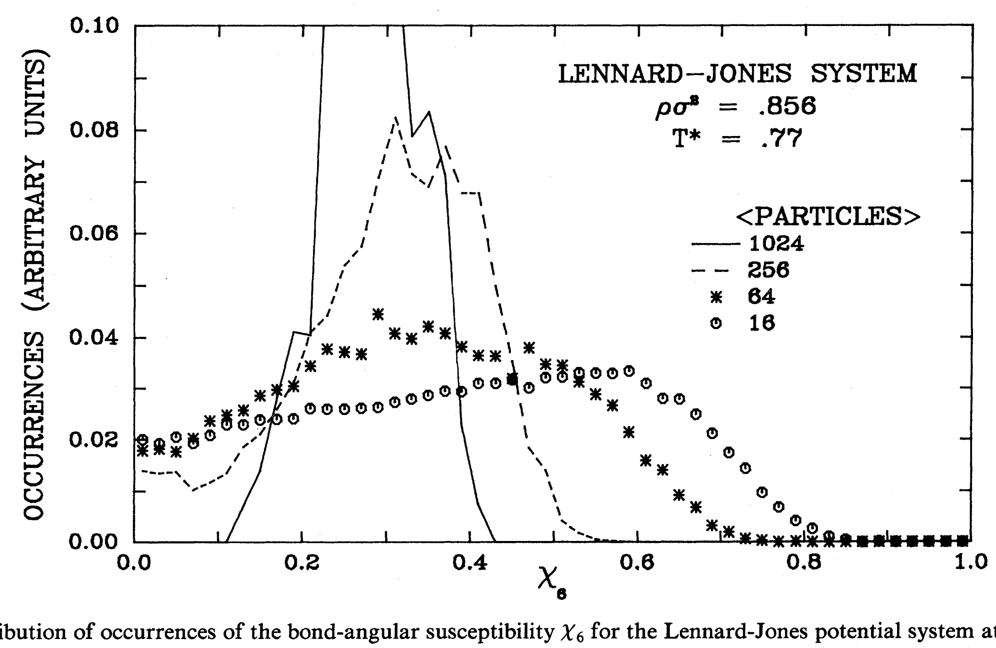

FIG. 2. Distribution of occurrences of the bond-angular susceptibility χ_6 for the Lennard-Jones potential system at $\rho\sigma^2=0.856$ and $T^*=0.77$. Data are displayed for subsystems containing an average of 1024, 256, 64, and 16 particles.

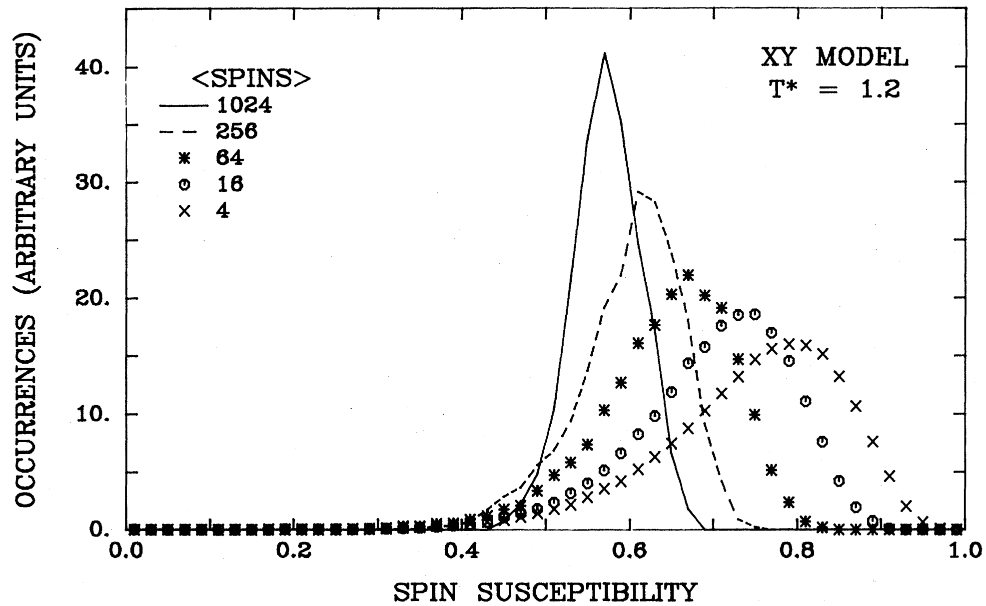

FIG. 3. Distribution of occurrences of spin susceptibility for the triangular XY model at $T^*=1.2$. Data are displayed for subsystems containing 1024, 256, 64, 16, and 4 spins.

The distributions for the solid just before melting and the liquid just before freezing for the hard-disk system are shown in Fig. 4. The behavior of the solid-phase distributions is as might be expected. A well-defined peak at a fairly large value of χ_6 broadens and shifts slightly as the subsystem size is decreased. In the fluid the distributions are peaked near zero as expected but are surprisingly broad. This broadening must be due to fluctuations giving rise to local regions with a high degree of bond-angular order. This observation is consistent with the large values of angular correlation length which have been observed in the fluid near freezing[7] and suggest that the melting transition is very weakly first order, showing large pretransitional fluctuations in angular order.

The results of combining the solid and fluid distributions in the appropriate proportions are compared with the data for hard disks at $\rho\sigma^2=0.891$ and 0.904 in Fig. 5 for subsystems containing an average of 16 particles. Our model of this distribution as a combination of liquid and solid distributions is not entirely adequate because for any

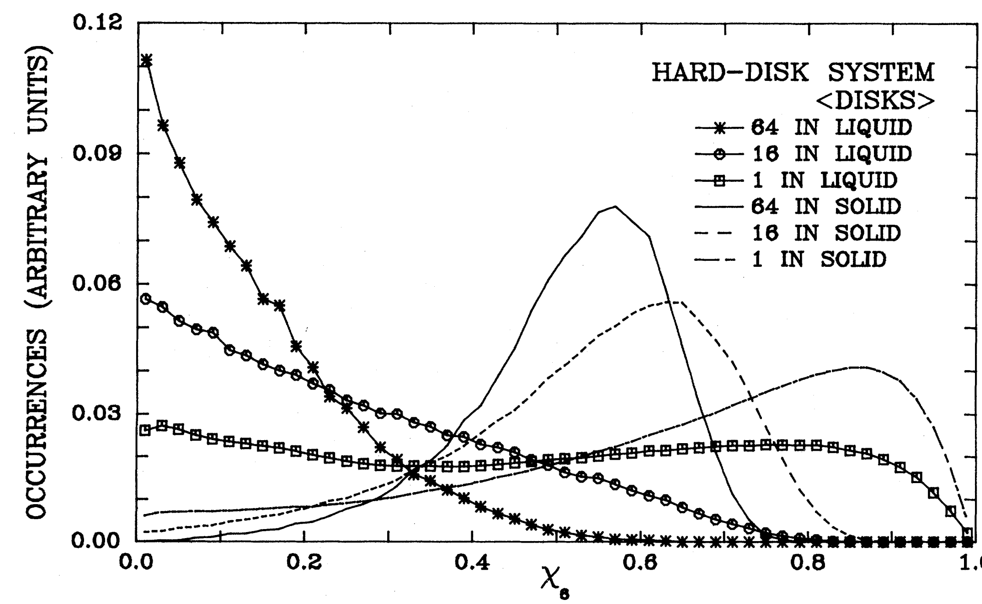

FIG. 4. Occurrences of bond-angular susceptibility χ_6 for hard disks in the fluid before freezing ($\rho\sigma^2=0.879$) and in the solid before melting ($\rho\sigma^2=0.910$).

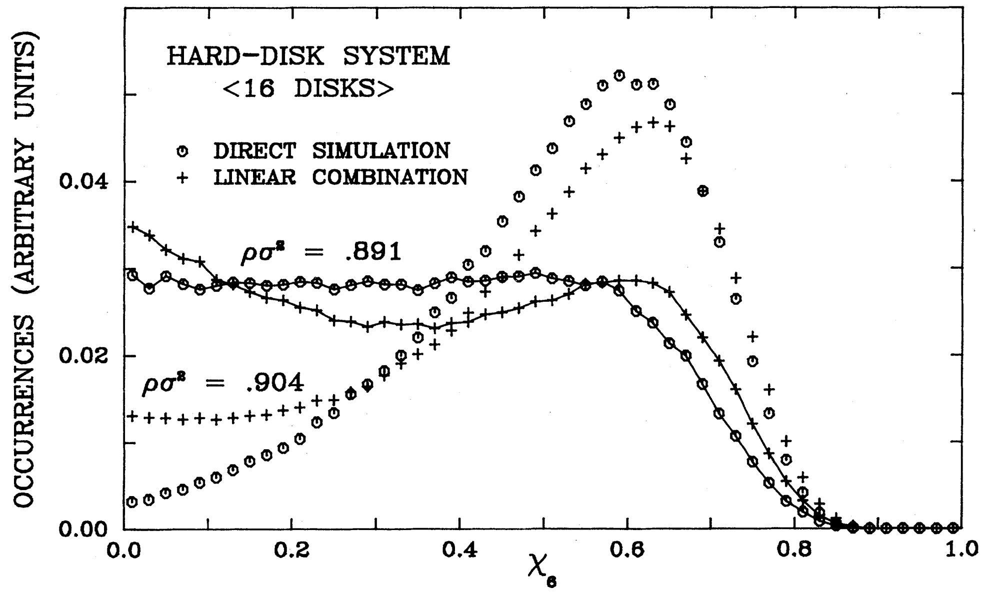

FIG. 5. Comparison of Monte Carlo data for subsystems containing an average of 16 particles in the intermediate region with combinations of fluid and solid distributions. The proportions of fluid to solid are 3:2 and 1:4 for $\rho\sigma^2=0.891$ and 0.904, respectively.

size of subsystem there will be some subsystems which span the boundary between regions of solid and fluid. These subsystems will produce a "filling in" of the middle of the distribution for χ_6 which will become more important as the size of the subsystems increases. This effect may be seen in Fig. 5.

If, however, we consider the distributions of values of χ_6 for individual particles, the combination of fluid and solid distributions should match the result in a two-phase region exactly. The results of this experiment are shown in Fig. 6. The agreement between the intermediate-region data and the coexistence model is extremely good on this level. Referring again to Fig. 5 we see that the agreement is also quite good for subsystems containing an average of 16 particles at both densities shown. The size of subsystem at which the model no longer works is a measure of the size of the patches of liquid and solid present in the coexistence region. From these results we would estimate a size of about 50 particles, roughly consistent with previous estimates of about 100 particles obtained from particle trajectory plots.[3]

We thus conclude from this investigation that the melt-

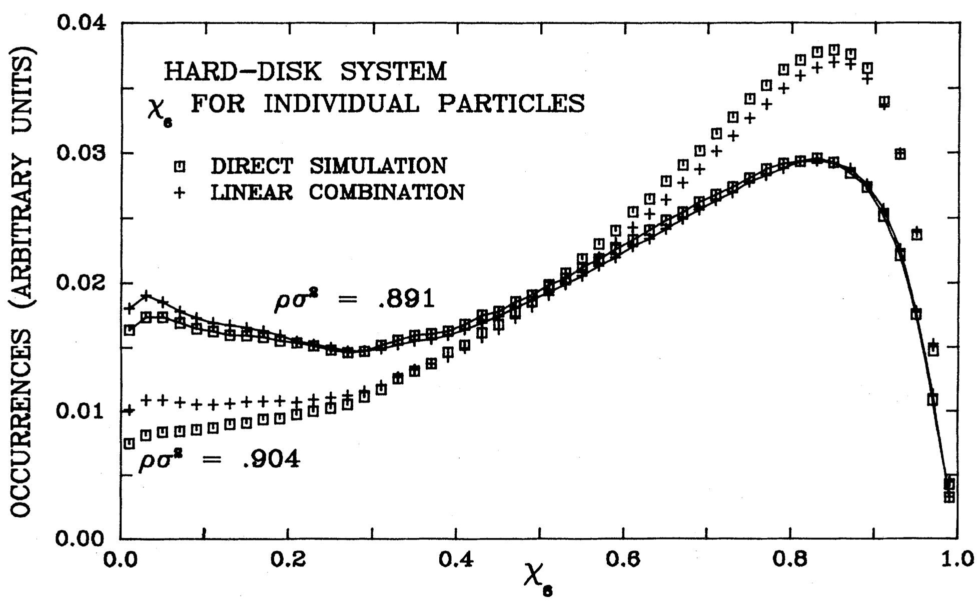

FIG. 6. Comparison of Monte Carlo data for χ_6 for individual particles with combinations of fluid and solid distributions. The proportions of fluid to solid are as in Fig. 5.

ing transition of 1024 Lennard-Jones particles or hard disks is first order and displays a region of two-phase coexistence. Our results are inconsistent with the identification of the intermediate region as a hexatic phase. They are also inconsistent with the previous conjectures that the melting transition of the Lennard-Jones potential system at densities such as 0.856 might be qualitatively different from that at high density.[3] These findings are consistent with the study of Weeks and Broughton showing that the Lennard-Jones potential system may be modeled well by a repulsive force potential.[8]

Although our results are conclusive for our finite system with periodic boundary conditions, we cannot rule out the possibility that these constraints have forced a transition which would be continuous in the thermodynamic limit to appear as a weak first-order transition. Indeed, our results confirm the existence of significant fluctuations of angular order prior to freezing. Studies of size dependence of this transition are certainly of great importance. Recent work by Bakker, Bruin, and Hilhorst on a system of 10 000 particles is a step in this direction.[7] We believe that the technique described in this paper should be useful in determining the effect of changes in size on the nature of the melting transition.

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*Present address: Department of Physics, Carnegie-Mellon University, Pittsburgh PA 15213.

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